

Robust Self-Distribution of Homogeneous Robotic Teams with Higher-Order Dynamics and without Initial Informational Connectivity^{*}

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Abstract—A homogeneous team of an unknown size consists of anonymous robots that carry no communication facilities and cannot play distinct roles and so should be driven by a common control rule. Robots with second-order dynamics, elements of higher-order ones, uncertainties, and limited speeds and control inputs are handled. The state of any robot is partly characterized by a scalar variable. Every robot has access to its own speed and the relative value of this variable attributed to any peer that is within a given “visibility distance” and in the clear line of sight. A control protocol is proposed for self-deployment of the team into nodes of a segment of a uniform lattice with a given spacing in three scenarios: 1) the team should eventually halt, 2) an end point of the segment is pre-specified and moves, 3) the segment should incorporate a given moving object. The protocol does not use the neighbor’s velocities and in the case 2), does not need initial informational connectivity of the team. These findings are illustrated by design of a navigation strategy for building a virtual antenna array in 3D by robotic drones. The proposed algorithms are justified by rigorous global convergence results; their performance is confirmed by computer simulations.

Index Terms—Autonomous robots, multiagent coordination, distributed algorithms, stability analysis, global convergence

I. INTRODUCTION

The paper considers a team of anonymous mobile robots that use only local sensor data, cannot play distinct roles and, thereby, should be individually driven by a common control rule, but must self-organize without communication or external coordination. This problem lies at the crossroad of swarm robotics [1] and behavior-based AI [2], where the focus is on simple local rules resulting in a complex collective behavior. Whereas the above uniformity offers plenty of advantages regarding scalability and fault tolerance, it challenges most of conventional formation control approaches (see, e.g., [3] for a survey) that rely on, e.g., virtual structures or designated leaders [4], [5], role differentiation or pre-established communication links among the agents, their access to a portion of global information about the team’s topology, etc.

This paper deals with the team’s self-deployment into a fully and singly filled segment of an even 1D lattice. Specifically, the current condition of every robot is partly characterized by a scalar variable called *state*. The swarm of an unknown size should autonomously arrive at a situation where the states are evenly spaced with a given step. Three tasks are handled where, respectively, 1) the robots should eventually stop and the position of the segment does not matter, 2) an end of the segment should equal the state of a given external anonymous moving agent, 3) the segment should merely contain this state.

This problem has a key relevance to a whole range of applications, e.g., ad-hoc building virtual antenna arrays of transceivers or hydrophones by fully homogeneous (in particular, unnumbered and unable to establish leader-follower relations) swarms of aerial or underwater drones. If the lattice spacing is set to be zero, the tasks 1) and 2,3) reduce to achieving consensus and trajectory following, respectively. In general, they have some similarities with (but differ from) the problem of even self-deployment on an unknown line segment (whose origins can be traced back to J. Darboux [6]) or circle.

Existing techniques for multi-agent coordination relevant to our target scenario can be broadly put into the following categories, each with its own serious limitations.

Consensus Based Formation Control. Most solutions require either velocity measurements [7]–[9] or connectivity preservation, unlike our vision-only approach. Whereas some works address nonlinear dynamics, they typically allow for unbounded controls, which is impractical for physical robots.

Behavior-Based Approaches. Virtual force methods [10], [11] often lack convergence guarantees. Noteworthy exceptions exist for coverage tasks, but they rely on clock synchronization or use elements of centralized decision making.

Even Distribution over a Line Segment or Circle. These problems have been mostly studied in the case of single or double integrator kinematics [12]–[14] and for discrete-time systems [15], [16]. However, these studies i) typically ignore realistic second-order dynamics with uncertainties and higher-order components and ii) often deal with teams that are homogeneous only in some respects (not fully), e.g., have

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access to global numbering of the robots.

Information Flow Connectivity. Preserving existing links or overall initial connectivity is relatively well-studied [17]–[23], though the related protocols often assume unlimited velocities [23] or complex topology management with access to global information about it [18], [19]. To the best of the authors’ knowledge, creating connectivity from initially disconnected states remains a largely open issue.

In order to contribute to filling these gaps, the paper offers i) a sliding-mode control law for the case of uncertain second-order dynamics with higher-order components, requiring no access to the velocities of the partners, ii) mathematically rigorous convergence and collision avoidance guarantees with respecting a priori velocity and control bounds, iii) elaboration of the issue of emerging informational connectivity from initially disconnected states; iv) justification of the robustness to dropouts/insertions of agents during team’s operation.

The paper offers a completed and rigorous justification of the algorithm’s convergence, theoretical findings are confirmed and complemented via computer simulation tests, including those with realistic Webots simulator.

The body of the paper is organized as follows. Sec. II describes the problem setup and assumptions. Sec. III–V present solutions for the tasks 1), 3), and 2), respectively. Sec. VI reports on computer simulation tests and Sec. VII offers conclusions. Due to the paper length limitations, the proofs of the presented theoretical results will be given in its full version and are available upon request.

Throughout the paper, $[k_1 : k_2]$ is the set of integers j such that $k_1 \leq j \leq k_2$; $:=$ means “is defined to be”; $\|\vec{v}\|$ is the Euclidean norm of the vector $\vec{v}$; L_{loc}^∞ is the set of measurable functions essentially bounded on any finite time interval.

II. PROBLEM STATEMENT AND ASSUMPTIONS

We consider $N \geq 2$ robots that operate in continuous time t . They are labeled 0 through $N - 1$ for our convenience, but no robot has access to these labels. At time t , robot i is (partly) characterized by a scalar $y_i(t) \in \mathbb{R}$ (called *state*). In total, they form the team’s *state* $Y(t) := \{y_i(t)\}_{i=0}^{N-1} \in \mathbb{R}^N$, and the time derivatives $v_i := \dot{y}_i$ constitute the *vector of the speeds* $V(t) := \{v_i(t)\}_{i=0}^{N-1}$. Any robot i builds its own control input

$$u_i \in \Omega_i \subset \mathbb{R}^{k_i}, \quad (1)$$

where Ω_i is a set of admissible controls. The *speeds of the robots* $v_i := \dot{y}_i$ should obey a given upper bound $\bar{v} \in (0, +\infty]$:

$$|v_i(t)| < \bar{v} \quad \forall t, i. \quad (2)$$

We handle robots with an uncertain state dynamics described by ODE’s of the following form:

$$\ddot{y}_i = f_i[t, Y(t), V(t), u_i(t), \mu_i(t)]. \quad (3)$$

Here $\mu_i \in \mathbb{R}^{m_i}$ is an *exogenous input*. It is partly used to catch the effect from time-varying $\lambda(\cdot)$ and constant λ_0 uncertainties, which may be caused by, e.g., external disturbances or ambiguity in parameters. The uncertainties are limited by

$$\lambda(t) \in \Lambda \subset \mathbb{R}^k, \quad \lambda_0 \in \Lambda_0 \subset \mathbb{R}^{k_0}, \quad \lambda(\cdot) \in L_{\text{loc}}^\infty(0, \infty), \quad (4)$$

where Λ, Λ_0 are certain sets. It is possible that (3) represents only a part of a more complex r -th order dynamics, which involves an extra phase variable $z \in \mathbb{R}^p$ and an equation

$$\dot{z}^{(r)}(t) = \mathcal{Z}[t, Z(t), Y(t), V(t), U(t), \lambda(t), \lambda_0], \quad \text{where} \quad (5)$$

$$U(t) := \{u_i(t)\}_{i=0}^{N-1}, \quad Z(t) := \{z^{(k)}(t)\}_{k=0}^{r-1}. \quad (6)$$

The evolution equation (3) is affected by $z(\cdot), \lambda(\cdot), \lambda_0$ via μ_i :

$$\mu_i(t) = \mathcal{M}_i[t, Z(t), Y(t), V(t), \lambda(t), \lambda_0] \in \mathbb{R}^{m_i}. \quad (7)$$

It is tacitly assumed that the initial data for $z(\cdot)$ are determined from $\lambda_0, Y(0), V(0)$. Also, $\lambda(\cdot)$ may include control inputs intended to drive $z(\cdot)$ and unknown to any robot i .

Equations (5) and (7) may account for unmodeled dynamics, inter-influence of the robots, and other factors that are out of control by robot i . The functions $f_i(\cdot), \mathcal{Z}(\cdot), \mathcal{M}_i(\cdot)$ are locally Lipschitz continuous. These functions and the sets Λ, Λ_0 may be unknown; they are basically used to qualitatively illuminate interconnection and general traits of various factors.

Robot i is unaware of the team’s size N , (both current and initial) state Y and its derivative, and the quantities $\lambda(t), \lambda_0$. On the positive side, it measures its own speed $v_i(t)$ and the relative states of some peers j subject to the following.

Assumption 1: There is a constant $R_{\text{vis}} > 0$ such that any robot i has access to the relative state $y_j(t) - y_i(t)$ of any peer j for which $|y_i(t) - y_j(t)| < R_{\text{vis}}, y_j(t) \neq y_i(t)$, and there is no robot k whose state $y_k(t)$ lies between $y_i(t)$ and $y_j(t)$.

Such peers j are said to be *visible* to robot i , with adding *upward* and *downward* if $y_j(t) > y_i(t)$ and $y_j(t) < y_i(t)$, respectively. There is no more than one upward/downward visible peer, and may be none.

A harmful effect from the “intermediate” robots is illustrated by the situation where unmanned cars move on a road in a chain and assess the inter-vehicle distances by using front-view video cameras. Then a bulky car in front may block the view of anything that lies further on the road.

We pursue three objectives. They consist in driving the team into one of the following situations, respectively:

- 1) A *steady distribution with the step* $w \geq 0$. This means that there exists $y_* \in \mathbb{R}$ and an enumeration of the robots such that $y_i = y_* + iw$ for all i .
- 2) A *distribution with the step* $w \geq 0$ and a *lower (upper) dynamic anchor* $y_\dagger(t)$. This comes to 1) where y_* varies with time and $y_* := y_\dagger(t) + w$ ($y_* := y_\dagger(t) - Nw$).
- 3) A *distribution with the step* $w \geq 0$ and a *pacemaker* $y_\dagger(t)$. This comes to 1), where $y_*(t) := y_\dagger(t) + wk$ for some integer k , and the enumeration from 1) covers the pacemaker $y_\dagger$ and treats it on a par with the robots.

In **1)**, the value of y_* does not matter and the robots have no common notion of up and down regarding the states. If $w > 0$, the goal is to fully and singly fill some segment S of an even w -spaced lattice. With $w = 0$, the goal is to achieve consensus on the value of the state. In **2)**, an extreme node of S is given by $y_\dagger(t)$; and S should lie on one side to $y_\dagger(t)$. The robots are aware about this side but do not know $y_\dagger(t)$.

In **3**), the position of the lattice and its travel speed are set by the pacemaker, which participates in forming the segment S . Like in **1**), the robots have no common notion of up and down and none of them knows the direction to the pacemaker; otherwise, **3**) is reducible to the two copies of **2**) w.r.t. to the parts of the team that lie on different sides of the pacemaker.

In the cases **2**) and **3**), we introduce a fictitious robot, labeled with -1 , with the state $y_{\dagger}(t)$. Its perception by true robots is limited; see Asm. 2 below. With $w = 0$, **2**) and **3**) mean that all states $y_i(t)$ track a common and given trajectory $y_{\dagger}(t)$.

Definition 1: The team is said to *asymptotically achieve a distribution* of some of the types **1**)–**3**) if the states never collide $y_i(t) \neq y_j(t) \forall i \neq j, t$ (including the state y_{-1} in the cases **2**) and **3**)) and as $t \rightarrow \infty$, the following statements hold in the cases **1**) and **2,3**) (with $i \in [-1 : N - 1]$), respectively:

- **s.1)** the robots' speeds go to 0 and their states converge to steady points whose distribution meets **1**);
- **s.2)** $\Delta_{i,\dagger}(t) := y_i(t) - y_{\dagger}(t)$ goes to a steady point $\Delta_{i,\dagger}^\infty$ and $\Delta_{i,\dagger}(t) \rightarrow 0$, whereas the properties from the respective item **2**) or **3**) hold with $y_i(t) := y_{\dagger}(t) + \Delta_{i,\dagger}^\infty$.

A common rule is to be designed by which every robot generates its own control based on the available local data so that the robots are driven to a distribution of the chosen type.

Assumption 2: In the cases **2**) and **3**), Asm. 1 holds for the true robots even if the set of their peers is enriched with the fictitious robot. The true robots cannot distinguish between true and fictitious peers and have to treat them equally.

The attainability of the control objective falls under strong doubt if either the information connectivity of the team is broken at the targeted deployment or the uncertain and unmanageable factors cannot be overpowered by control actions. These situations are excluded by the following two assumptions.

Assumption 3: With the targeted placement, the adjacent robots "see" each other: $w < R_{\text{vis}}$.

Assumption 4: For any i , there exist two known admissible controls $u_i^{(+1)}, u_i^{(-1)} \in \Omega_i$ and a constant $e_{u,i} > 0$ such that

$$\pm f_i[t, Y, V, u_i^{(\pm 1)}, \mu_i(t)] \geq e_{u,i} \quad (8)$$

in any feasible circumstances, i.e., for any $i, t \geq 0, Y, V \in \mathbb{R}^N$, any μ_i given by (5) and (7), and any $\lambda(\cdot), \lambda_0$ satisfying (4).

These controls accelerate and decelerate the robot, respectively, under any possible situations. Also, $e_{u,i}$ may be small and is used to express that the concerned acceleration/deceleration does not degrade to zero as some arguments from (8) increase without limits (if possible).

Our last assumption excludes the situation where a solution of (3), (5), (7) cannot be continued ad-infinitum; this assumption may be replaced by any condition that ensures the same.

Assumption 5: For any i , there exist coefficients $A_i(\lambda_0) \geq 0, B_i(\lambda_0) \geq 0, q_i(\lambda_0) \geq 0$ such that the following estimates hold

$$\begin{aligned} & |f_i[t, Y, V, u_i, \mathcal{M}_i(t, Z, Y, V, \lambda, \lambda_0)]| + |\mathcal{Z}[t, Z, Y, V, U, \lambda, \lambda_0]| \\ & + |\mathcal{M}_i[t, Z, Y, V, \lambda, \lambda_0]| \\ & \leq A_i(\lambda_0) + B_i(\lambda_0) [\|Y\| + \|V\| + \|Z\| + \|\lambda\|^{q_i(\lambda_0)}] \end{aligned}$$

whenever $u_i = u_i^{(\pm 1)}, v_i \in [-\bar{v}, \bar{v}], t \geq 0, Y, V \in \mathbb{R}^N, \lambda \in \Lambda, \lambda_0 \in \Lambda_0$, and (4) is true.

We close the section with illustrations of scenarios where the above model applies.

e.1) The state equation is given by $\ddot{y}_i = u_i + M_i(t, y_i, \dot{y}_i, \lambda, \lambda_0)$, where $M_i(\cdot)$ is a Lipschitz continuous function of its arguments. An example is given by wheeled robots moving over a road. Then y_i describes the position of the i th robot on the road, u_i stands for its nominal driving acceleration, and $M_i(t, y_i, \dot{y}_i, \lambda, \lambda_0)$ accounts for the aberrations due to, e.g., volatility of the traction coefficient and roadway slope along the road, the effect of the aerodynamic drag force, and incomplete knowledge of the robot's parameters.

Suppose that the function $M_i(\cdot)$ is upper bounded by a constant $\bar{M}$, which is rather usual in practical setting. Then Asm. 5 holds and Asm. 4 means that $\pm u_i^{(\pm 1)} > \bar{M}$ for some $u_i^{(\pm 1)} \in \Omega_i$ and any i , or in other words, that robot i is able to overpower the exogenous factors by its own actuation effort.

e.2) Robot i is described by two ODE's: $m_i \ddot{y}_i = u_i + M_i(t, y_i, \dot{y}_i, m_i, \lambda, \lambda_0), \dot{m}_i = -K_i(t, y_i, \dot{y}_i, m_i, u_i, \lambda, \lambda_0) \in \mathbb{R}$, where the functions $M_i(\cdot)$ and $K_i(\cdot)$ are Lipschitz continuous and bounded. This example modifies e.1) and assumes robots with internal combustion engines; m_i is the vehicle's mass, which is not steady because of the fuel consumption. The first equation is similar in spirit to that from e.1), but now u_i and M_i have the meaning of forces. The second equation describes the fuel consumption, whose rate may depend on the vehicle's speed $\dot{y}_i$ and acceleration u_i/m_i [24].

It is easy to see that Asm. 5 still holds and Asm. 4 has the same meaning as in e.1).

III. STEADY DISTRIBUTION WITH THE STEP w

Prerequisites. Irrespective of the objective, our control rules use a smooth function $\Xi(d) \in \mathbb{R}$ of $d \in [0, R_{\text{vis}}]$ such that

$$\Xi'(d) > 0 \forall d \in [0, R_{\text{vis}}], \quad \Xi(0) = 0. \quad (9)$$

Examples include $\Xi(d) = kd, \Xi(d) = k \arctan d, \Xi(d) := \frac{kd}{1+d}$, where $k > 0$ is a free parameter. This function is to be chosen by the designer of the controller.

Data pre-processing. Given $\varsigma \geq 0$, the *upper* $g_i^+(t|\varsigma) \geq 0$ and *lower* $g_i^-(t|\varsigma) \geq 0$ ς -gap of robot i are defined as

$$g_i^+(t|\varsigma) := \begin{cases} y_j(t) - y_i(t) & \text{if } \left| \begin{array}{l} \text{there is an upward} \\ \text{visible peer } j \end{array} \right| \\ \varsigma & \text{otherwise,} \end{cases} \quad (10)$$

$$g_i^-(t|\varsigma) := \begin{cases} y_i(t) - y_j(t) & \text{if } \left| \begin{array}{l} \text{there is a downward} \\ \text{visible peer } j \end{array} \right| \\ \varsigma & \text{otherwise.} \end{cases} \quad (11)$$

Control rule 1. The control input to robot i is formed as

$$u_i(t) := u_i^{(-\text{sgn } \Sigma_i(t))}, \quad \text{where} \quad (12)$$

$$\Sigma_i(t) := v_i(t) - \Xi[g_i^+(t|w)] + \Xi[g_i^-(t|w)], \quad (13)$$

and $u_i^{(\pm 1)}$ are taken from Asm. 4. The input (12) clearly meets (1). In closed loop and for given uncertainties, the state $X := (Y, \dot{Y}, Z)$ obeys an ODE $\dot{X} = F(t, X)$ with a discontinuous r.h.s. We treat its solution in the Filippov's sense [25].

Remark 1: Given $\tau \geq 0$, $\lambda(\cdot) \in L_{\text{loc}}^\infty([\tau, \infty) \rightarrow \Lambda)$, $\lambda_0 \in \Lambda_0$, Y_0 , and V_0 , there is a solution of the closed-loop ODE that starts with Y_0, V_0 at $t = \tau$ and is defined for all $t \geq \tau$.

We consider all solutions except for the inviable ones. They are those that go over a repelling discontinuity surface.

Remark 2: The control rule 1 is conformable to the anonymity of the data.

This holds since re-enumeration of the robots does not affect the quantities (10), (11) and so the control (12).

To proceed, we need some definitions. Let $\Gamma(t)$ denote the undirected graph whose nodes correspond to the robots (and are marked with the same labels) and in which nodes i and j are linked if and only if the robots $i \neq j$ are mutually visible.

Definition 2: A sequence $\{y_i\}$ is said to be *non-breaking* if the related graph Γ is connected, and *collision-free* if $y_i \neq y_j \forall i \neq j$. Based on Asm. 4, we introduce a constant $e_u > 0$ such that

$$0 < e_u \leq \min_{i \in [0:N-1]} e_{u,i}. \quad (14)$$

Theorem 1: Let (9) and Asm. 1, 3–5 hold, and the robots be driven by the control rule 1 and initially obey (2):

$$|\dot{y}_i(0)| < \bar{v} \quad \forall i. \quad (15)$$

Suppose that the initial state of the team is non-breaking and the function $\Xi(\cdot)$ in (12) is chosen so that

$$\bar{\Xi} := \Xi(R_{\text{vis}}) < \bar{v}, \quad \bar{\Xi}' := \max_{d \in [0, R_{\text{vis}}]} \Xi'(d) < \frac{e_u}{4\bar{v}}. \quad (16)$$

Let the initial deployment of the robots be collision-free and

$$-\Xi[g_i^-(0|w)] \leq v_i(0) \leq \Xi[g_i^+(0|w)], \quad (17)$$

$$\Xi[g_i^+(0|w)] - \bar{\Xi} \leq v_i(0) \leq -\Xi[g_i^-(0|w)] + \bar{\Xi}. \quad (18)$$

Then the following statements hold:

- i) The team asymptotically achieves a steady distribution with the step w in the sense of Defn. 1;
- ii) Their deployment remains non-breaking;
- iii) Inequalities (17), (18) are true for any time and robot:

$$-\Xi[g_i^-(t|w)] \leq v_i(t) \leq \Xi[g_i^+(t|w)], \quad (19)$$

$$\Xi[g_i^+(t|w)] - \bar{\Xi} \leq v_i(t) \leq -\Xi[g_i^-(t|w)] + \bar{\Xi}. \quad (20)$$

Inequalities (16) give guidelines for choosing $\Xi(\cdot)$ and are met with all small enough functions $\Xi(\cdot)$ with a small enough slope. By (9), inequalities (17) and (18) are satisfied if the robots are initially steady $v_i(0) = 0 \forall i$. In general, they require that the initial speeds are not excessively large relative to 1) separation among the robots and 2) reserve until breaking the sensorial contact between them: otherwise, inter-robot collisions or loss of informational connectivity of the team may be inevitable due to inertia and the constraints (1).

Remark 3: Unlike most of second-degree consensus protocols and their offsprings to formation control, the proposed control rule does not need access to the velocities of the peers.

Remark 4: Let the assumptions of Thm. 1 hold and some robots disappear during the mission. Suppose that just after any such event, the deployment of the rest of the team remains non-breaking and (20) holds for all its members. Then i)–iii) from Thm. 1 are true for the remaining team.

This follows from Thm. 1 via consecutively shifting the zero-of-time to the times of the losses. The requirement to the team to remain non-breaking means that whenever two remaining robots instantaneously loose all “intermediate” partners, these two robots should be within the visibility range of each other. By (9) and (16), $\Xi[d] - \bar{\Xi} < 0$ for all $d \in [0, R_{\text{vis}})$. So to keep (20) valid, as is assumed in Rem. 4, the speeds of the both remaining robots should be small enough. By Defn. 1, they tend to zero and $g_i^\pm(t|w)$ go to w as $t \rightarrow \infty$; so (20) is kept valid if the losses occur late enough.

Remark 5: Let the assumptions of Thm. 1 hold and extra robots join the team during its operation. Suppose that just after their appearance, the state of the united team remains collision-free and inequalities (19), (20) hold for this team. Then i)–iii) from Thm. 1 are true for the united team.

This is justified by consecutively shifting the zero-of-time to the moments of the admissions and applying Thm. 1.

In Rem. 5, the assumption related to (19) and (20) means that any newcomer should join the team with small enough speed. For any “oldie” i , (19) and (20) should be checked only if a “newcomer” becomes visible to i . Then (20) holds for sure, whereas retaining (19) means that the newcomer should not appear in a nonproportionally short distance.

Remark 6: The control rule 1 does not need that all robots have a common view on the direction of the state ascending.

Indeed, let a particular robot i reverses its opinion. Then in its reference frame, $f_i(\dots) \mapsto -f_i(\dots)$ in (3) and so $u_i^{(\pm 1)} \mapsto u_i^{(\mp 1)}$ in Asm. 4, $v_i \mapsto -v_i$, $g_i^\pm(\cdot|w) \mapsto g_i^\mp(\cdot|w)$ by (10) and (11), $\Sigma_i \mapsto -\Sigma_i$ by (13), and so formula (12) produces just the same control input as before the change of the opinion.

IV. EVEN SELF-DISTRIBUTION WITH A PACEMAKER

Control rule 2 is run on any true robot i and is given by (12) and (13), where the fictitious robot -1 (the pacemaker) with the state $y_{\dagger}(t)$ is taken into account in (10), (11). The *extended team* is the collections of the true robots plus this fictitious one. Rem. 1–3 evidently extend to the case at hands.

Theorem 2: Suppose that (9) and Asm. 2–5 hold and the speed of the pacemaker and its acceleration are bounded

$$\bar{v}_{\dagger} := \sup_{t \geq 0} |\dot{y}_{\dagger}(t)| < \infty, \quad \bar{a}_{\dagger} := \sup_{t \geq 0} |\ddot{y}_{\dagger}(t)| < \infty. \quad (21)$$

Suppose also that any true robot $i \in [0 : N-1]$ meets (17), (18), where $v_i(0) - v_{\dagger}(0)$ is put in place of $v_i(0)$, as well as the following respective enhancements of (15) and (8) hold:

$$|\dot{y}_i(0)| < \bar{v} - \bar{v}_{\dagger}, \quad e_u := \min_{i \in [0:N-1]} e_{u,i} > \bar{a}_{\dagger}. \quad (22)$$

Let (16) be true with $\bar{v} := \bar{v} - \bar{v}_{\dagger}$, $e_u := e_u - \bar{a}_{\dagger} > 0$ and initially the state of the extended team be collision-free and non-breaking. Then the following assertions are valid:

- i) The true robots achieve a distribution with the step $w \geq 0$ and a pacemaker $y_{\dagger}(t)$ in the sense of Defn. 1;
- ii) The state of the extended team is always non-breaking;
- iii) For any true robot i , (19), (20) hold at any time $t \geq 0$ provided that $v_i(t) - v_{\dagger}(t)$ is put in place of $v_i(t)$.

Rem. 4–6 remain valid in the case at hands (provided that (19), (20) are modified as is delineated in iii) of Thm. 2).

V. THE CASE OF A DYNAMIC ANCHOR $y_{\dagger} = y_{\dagger}(t)$ AND NO INITIAL CONNECTIVITY OF THE TEAM

We focus on the situation of a lower anchor and assume that its speed and acceleration are bounded, i.e., (21) holds.

Control rule 3 is still given by (12), where $\Sigma_i(\cdot)$ is, however, defined a somewhat differently:

$$\Sigma_i(t) := v_i(t) - \Xi[g_i^+(t|w)] + \Xi[g_i^-(t|R_{\text{vis}})], \quad (23)$$

$$\text{where } g_i^{\pm}(t|w) := \min \{g_i^{\pm}(t|R_{\text{vis}}); w\}. \quad (24)$$

Here $g_i^{\pm}(t|R_{\text{vis}})$ are defined by (10) and (11), where the fictitious robot “-1” with the state $y_{\dagger}(t)$ is taken into account.

Control rule 3 assumes that all robots have access to a common direction of the state ascending and the anchor should be lower w.r.t. exactly this direction.

Rem. 1–3 evidently extend to the case at hands.

Theorem 3: Let (9), (21), (22), and Asm. 2–5 hold and the following analog of (17) be true for all $i \in [0 : N - 1]$:

$$-\Xi[g_i^-(0|R_{\text{vis}})] \leq v_i(0) - \dot{y}_{\dagger}(0) \leq \Xi[g_i^+(0|w)]. \quad (25)$$

Suppose that (16) holds with $\bar{v} := \bar{v} - \bar{v}_{\dagger}$, $e_u := e_u - \bar{a}_{\dagger} > 0$ and initially the team’s deployment is collision-free and is above the anchor. Then the following assertions hold:

- i) The team asymptotically achieves a distribution with the step w and the lower dynamic anchor $y_{\dagger}$;
- ii) Its deployment is always collision-free and above $y_{\dagger}$;
- iii) Inequalities (25) are true for any time and robot:

$$-\Xi[g_i^-(t|R_{\text{vis}})] \leq v_i(t) - \dot{y}_{\dagger}(t) \leq \Xi[g_i^+(t|w)]. \quad (26)$$

Remark 7: The conclusion of Thm. 3 holds even if initially some or all inter-robot gaps exceed R_{vis} to the point of being arbitrarily large, and so the team lacks informational connectivity. The thing is that sooner or later, this connectivity is created from nothing by the employed control rule.

It is also easy to see that the following analogs of Rem. 4 and 5 are true. These analogs are justified in a similar way, with referring to Thm. 3 instead of Thm. 1.

Remark 8: Let the assumptions of Thm. 3 hold and some robots disappear during the mission. Then i)–iii) from Thm. 3 are true for the remainder of the team.

Unlike Rem. 4, it is not needed that the losses of the robots do not violate the informational connectivity of the teams.

Remark 9: Suppose that the assumptions of Thm. 3 hold and extra robots join the team during its operation. Let just after their appearance, inequalities (26) hold for the united team and the state of any “newcomer” differs from those of all other robots and lies above the anchor. Then i)–iii) from Thm. 3 are true for the united team.

The material of this section almost literally extends on the situation of an upper anchor.

VI. COMPUTER SIMULATION TESTS: BUILDING VIRTUAL ANTENNA ARRAYS IN 3D.

A fully homogeneous swarm of N anonymous mobile robots operates in 3D. The robots should self-organize into a vertically aligned array above a certain unsteady location $\mathbf{p} = \mathbf{p}(t)$ (called *beacon*), whose altitude is constant and taken as zero. In the array, the distance gap between the neighboring robots must equal w_a and the altitude of the lowest robot must equal h , where h, w_a are given and $h \geq w_a > 0$.

The robots have access to the vertical and a common horizontal (e.g., to the north) directions, and so to the coordinate χ, y, z directions of a common Cartesian frame $\mathcal{F}$ rooted at $\mathbf{p}$, with the z -axis going vertically and upwards. In its local frame, every robot is able to pinpoint the relative location of the beacon or partner if the line of sight is not blocked and a given sensing distance R_{sen} is not exceeded, where $R_{\text{sen}} > w_a, h$. Any robot measures its altitude.

To build the control rule, we divide $\bar{u}$ and R_{sen} among the axes of $\mathcal{F}$ by picking $\bar{u}_b, R_b > 0$ for $b = \chi, y, z$ so that $\sum_b \bar{u}_b^2 \leq \bar{u}^2$, $\sum_b R_b^2 \leq R_{\text{sen}}^2$, and $R_z > w_a, h$. This is feasible since $R_{\text{sen}} > w_a, h$. For $b = \chi, y, z$, a mapping Ξ_b from $[0, R_b]$ to $[0, \bar{u}_b]$ is picked s.t. (9) holds with $R_{\text{vis}} := R_b$. The control u_i is formed in terms of its coordinates $u_i^x(t), u_i^y(t), u_i^z(t)$.

For $b = \chi, y$, the coordinate $u_i^b(t)$ is built via application of the control rule 2 with $\Xi := \Xi_b, w := 0$ and $R_{\text{vis}} := R_b$ to the relative coordinates $b_j(t) - b_i(t)$ of all visible objects j , including $\mathbf{p}$ (if applicable). The coordinate $u_i^z(t)$ is built via application of the control rule 3 with $\Xi := \Xi_z, w := w_a$ and $R_{\text{vis}} := R_z$ to the relative coordinates $z_j(t) - z_i(t)$ of all visible objects j . However, the coordinate of the beacon (if applicable) is preliminarily multiplied by w_a/h .

In the first round of the tests, point-mass robots with second-order integrator dynamics are considered: $\ddot{\mathbf{r}}_i = \mathbf{u}_i \in \mathbb{R}^3, \|\mathbf{u}_i\| \leq \bar{u}$. Here $\mathbf{r}_i$ is the location of robot i , and $\bar{u} > 0$ is given. The following assertion is immediate from Thm. 2, 3.

Proposition 1: Let the robots be initially at distinct altitudes $z_i(0) > h_{\dagger} - w_a$, motionless $\dot{\mathbf{r}}_i(0) = 0$, and close to one another in horizontal projection $|\chi_i(0)| < R_x/2, |y_i(0)| < R_y/2$. Then the robots do not collide with one another, never go below the zero altitude, and asymptotically form the required array above the beacon: as $t \rightarrow \infty$, the discrepancies between their horizontal projections and that of $\mathbf{p}$ go to zero and $z_i(t) \rightarrow h + iw_a \forall i$ under proper enumeration of the robots.

Here the altitudinal gaps between the robots may initially be so large that the information flow graph is edge-less.

In the first round of our tests, $w_a = 0.5$ m, $R_{\text{vis}} = 2.5$ m, $\bar{u} = 10.0$ m/sec, $h = 1.5w_a$, $\Xi(d) := \tanh(d)$, the control is updated with a step of 20 ms, the duration of the experiment equals 10000 time steps. To show that the requirements to the initial deployment from Prop. 1 are sufficient but not necessary, this deployment is formed randomly as follows

$$\begin{cases} \chi_{i+1} = \chi_i + r_i \sin \beta_i, \\ y_{i+1} = y_i - r_i \sin \alpha_i \cos \beta_i, \\ z_{i+1} = z_i + r_i \cos \alpha_i \cos \beta_i \end{cases} \quad i \in [-1, N - 1].$$

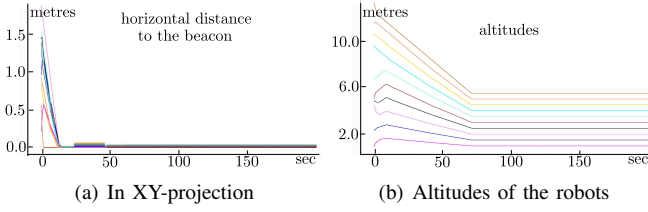


Fig. 1. The convergence process in the case of a single static anchor.

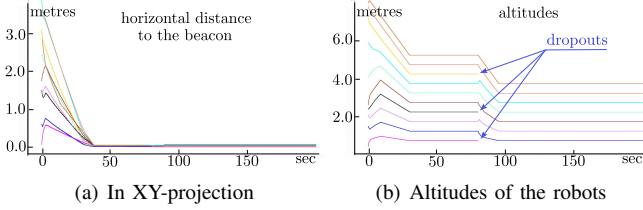


Fig. 2. The convergence process in the case of a linearly moving anchor.

Here x_{-1}, y_{-1}, z_{-1} are the initial coordinates of the beacon, whereas r_i, α_i, β_i are independently drawn from the uniform distribution over $[R_-, R_{\text{vis}}]$ and the normal distributions $\mathcal{N}(0, \sigma_\alpha), \mathcal{N}(0, \sigma_\beta)$, respectively. In our tests, $R_- = 0.1$ m and $\sigma_\alpha = \sigma_\beta = 0.6$, so there is a probability of breaching the afore-mentioned requirements.

For every simulation scenario, many initial deployments are generated; among them, a typical result is discussed. For each scenario, animation of the associated 3D maneuver of the team is available at <https://tinyurl.com/4vz348xn>, where the robots and beacon are shown as small multicolored points and a bold red dot, respectively. The graphs in the subsequent figures provide more detailed quantitative data on the convergence and performance of the algorithm; differently colored graphs are related to different robots.

Fig. 1 addresses a scenario where the beacon is static. As can be seen, a group of 10 robots starts higher the beacon, but not strictly above it. By Fig. 1(a), the robots align themselves in a nearly perfect vertical line above the beacon as quickly as in ≈ 10 sec. By Fig. 1(b), it takes ≈ 70 sec to arrive at the targeted altitudinal distribution. The main part of this transient (from ≈ 10 sec to ≈ 70 sec) is monotonic and linear.

In the scenario from Fig. 2, the beacon slowly moves along the x -axis with a constant speed of 0.05 m/sec. In addition to moving beacon, the team faces one more challenge: three robots are randomly dropped during the mission. Like in the previous experiment, the team reaches the desired even alignment after a short transient period and then moves strictly above the beacon, while forming a regular antenna array. Disappearance of three robots at ≈ 80 sec does not cause any visible deviation of the horizontal projections of the remaining robots from the beacon but expectedly damages the even altitudinal distribution. However, after another short transient, the reduced team restores such a distribution with fewer robots and the same altitudinal gap, as is required.

Fig. 3 addresses the case where the robotic team has to take on other challenges. First, the beacon perpetually

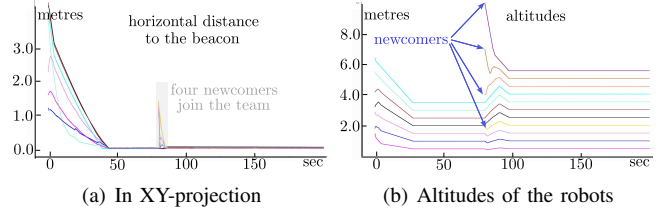


Fig. 3. The beacon moves over a circle and the team admits newcomers.

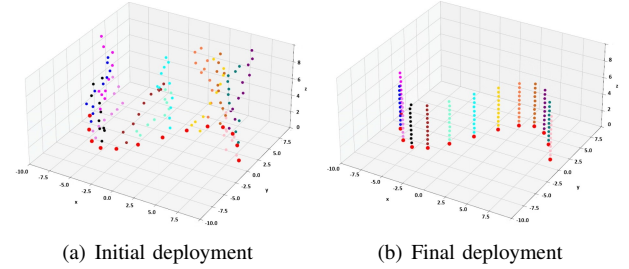


Fig. 4. Several static beacons, each with its own family of robots.

changes direction of motion by traveling with a speed of 0.02 m/sec over a horizontal circle with a radius of 4.0 m. Second, whereas initially the team consists of 7 robots, four extra robots suddenly join the team, starting with occasional locations. As follows from Fig. 3(a), they are horizontally far from the beacon; Fig. 3(b) shows that their admission ruins the previously achieved even altitudinal distribution. From the respective moment on, both “newcomers” and “oldies” run the algorithm as usual, with taking into account one another if applicable. By Fig. 3, it takes only a few seconds for the united team to fix the above mismatches with the targeted deployment by creating a renewed evenly distributed vertical formation moving above the beacon.

Thus the algorithm copes with the mission despite continual displacement, both linear and curvilinear, of the beacon and unexpected changes in the team’s membership.

The goal of the next two experiments is to test the principal possibility to use the proposed control paradigm as an algorithmic element for building 2D arrays. In Fig. 4, thirteen static beacons are evenly distributed over a segment of a horizontal sinusoid. Every beacon is a “head” of a family of 10 robots, different families are disjoint. Any robot runs the proposed control rule with taking into account only members of its own family. According to Fig. 4, the algorithm successfully drives the robotic team from a fairly chaotic initial deployment into a properly organized 2D array.

The purpose of the second round of experiments is to preliminarily assess real world applicability of the proposed control paradigm. To this end, the Crazyflie quadcopter model was run in the Webots simulation environment <https://cyberbotics.com/>. Like before, the proposed control law is used to generate the forces along the axes of a Cartesian scene frame. Then a bank of PD and PID regulators transforms these forces into input control signals consistent with the interface of the Crazyflie model. These regulators are based on a

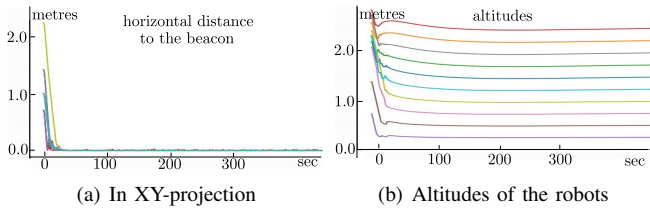


Fig. 5. The case of a static anchor: Webots simulation

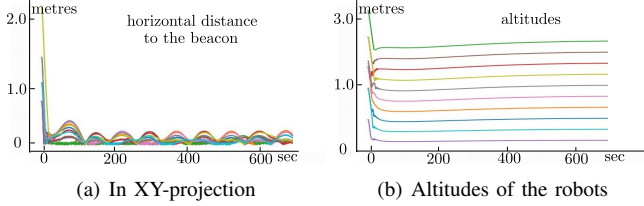


Fig. 6. The case of a moving anchor: Webots simulation

simple model of quadcopter flight dynamics and conventional hints of practical engineering.¹

Fig. 5 addresses a scenario where the (black) beacon is horizontally motionless. Ten robots start so that the horizontal projections are nearly indistinguishable for four groups of them in Fig. 5(a). According to Fig. 5(a) and 5(b), the desired antenna array is built at $t \approx 25$ sec; then it is stably maintained.

In Fig. 6, the team is additionally challenged by a horizontal motion of the beacon at a constant speed of 0.03 m/s along a circular path with a radius of 0.3 m. As can be seen, this does not visibly affect the quality of vertical distribution but degrades the accuracy of horizontal positioning, though it is still within a rather satisfactory bound of ≈ 0.3 m.

VII. CONCLUSIONS AND FUTURE WORK

A sliding-mode and computationally efficient distributed algorithm was presented that drives a fully homogeneous team of anonymous silent mobile robots into an uniform lattice with a given spacing. Robots with complex second-order dynamics, uncertainties, a priori limitations on the speeds and control inputs, and access to only local data were treated.

The algorithm was proven to provide theoretically precise convergence to the targeted deployment, to exclude inter-robots collisions, and to be robust against uncertainties, as well as unplanned losses and insertions of team members. More precisely, natural and partly inevitable conditions were elaborated under which these traits do hold. In some scenarios, the algorithm creates informational connectivity from nothing. Its performance and applicability was justified by global convergence theorems and confirmed by simulation tests.

Future work involves extension of the presented findings to the case of higher-degree outputs and specific systems, as well as studies on the issue of possible chattering and respective counter-measures, including higher-order sliding modes.

¹The details are omitted due to the paper length limitations.

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